PhD Talks Schedule 🧑‍🎓

Day 2 — Tuesday, April 8

09:15-09:30 ***Eric Legler:*** Contextual influences on habitual control through repetition bias: empirical evidence from a sequential decision-making task

09:30-09:45 ***Laís Muntini:*** The Environment as a Factor: Investigating the impact of the environment on cognition with VR

09:45-10:00 ***Caroline Stankozi:*** Context as interactive, dynamic, and perspectival

10:00-10:15 **BREAK**

10:15-10:30 ***Ekaterina Varkentin:*** Visual Narrative Comprehension in the Context of Aging and Stress

10:30-10:45 ***Simge Hamaloğlu:*** Perceiving the Unseen: How Causality Impacts Accuracy Without Enhancing Metacognitive Insight

10:45-11:00 ***Alexander Hölken:*** Mental Constraints - Naturalizing Mental Causation through Dynamical Systems Theory

Day 3 — Wednesday, April 9

09:00-09:15 ***Hassane Kissane:*** Neural and computational Modeling of Verb-Particle combinations

09:15-09:30 ***Max Mittenbühler:*** A Rational Trade-Off Between the Costs and Benefits of Automatic and Controlled Processing

09:30-09:45 ***Chang Liu:*** Investigating Speech Processing Load in a Conversational Context Task with Fish-Speech 1.5

09:45-10:00 ***Yijing (Quinn) Lin:*** Priming Effects on Verbal Analogy Performance Across IQ and Working Memory in Humans and AI

10:00-10:15 ***Alexandra Witt:*** Better together: flexibly biased learning rates in social learning

10:15-11:00 **BREAK**

Day 3 — Wednesday, April 9

11:00-11:15 ***Wenjia Xu:*** A closer look at the agent advantage effect: The Impact of Biting Orientations

11:15-11:30 ***Hossam Khalil:*** Human as an Asset - Building Human Holons for Human-Centred Production

11:30-11:45 ***Maedeh Amini:*** Investigating Social and Emotional Cognition in Children Using Face Processing in EEG Studies: A Systematic Review

Eric Legler: Contextual influences on habitual control through repetition bias: empirical evidence from a sequential decision-making task

Abstract: Human decisions are guided by the interplay between goal-directed and habitual mechanisms, with frequent repetition of actions leading to an increase in habitual control (Wood & Rünger, 2016). This increase in habitual control can be formalized with a frequency-based repetition bias, whereby each repetition of an action increases the probability of selecting that action again in the future. Notably, the repetition bias is value-free yet goal-aligned, as actions associated with high rewards are more likely to be selected. Recently, using a novel sequential decision-making task and computational modeling, we found evidence for such a repetition bias (Legler et al., 2024), based on the Bayesian contextual control model (Schwöbel et al., 2021).

In addition to being repetition-dependent, habitual control has been found to be context-dependent (Buabang et al., 2025; Wood & Rünger, 2016). However, studies that simultaneously investigate the combined influence of repetition and context on human decision making are surprisingly scarce (Hagger et al., 2023).

The present study aims to experimentally test the hypothesis that the repetition bias is context-dependent. To this end, we have modified our sequential decision-making task by introducing contexts in which different sequences of actions maximize rewards. We will compare cognitive models with and without a context-dependent repetition bias, to test whether participants use individual repetition biases for different contexts. This approach could be further used to investigate factors that facilitate or hinder the transfer of a learned repetition bias into new contexts.

References:

Buabang, E. K., Donegan, K. R., Rafei, P., & Gillan, C. M. (2025). Leveraging cognitive neuroscience for making and breaking real-world habits. Trends in Cognitive Sciences, 29(1), 41–59. https://doi.org/10.1016/j.tics.2024.10.006

Hagger, M. S., Hamilton, K., Phipps, D. J., Protogerou, C., Zhang, C.-Q., Girelli, L., Mallia, L., & Lucidi, F. (2023). Effects of habit and intention on behavior: Meta-analysis and test of key moderators. Motivation Science, 9(2), 73–94. https://doi.org/10.1037/mot0000294

Legler, E., Rivera, D. C., Schwöbel, S., Wagner, B. J., & Kiebel, S. (2024). Cognitive Computational Model Reveals Repetition Bias in a Sequential Decision-Making Task. bioRxiv 2024.05.30.596605. https://doi.org/10.1101/2024.05.30.596605

Schwöbel, S., Marković, D., Smolka, M. N., & Kiebel, S. J. (2021). Balancing control: A Bayesian interpretation of habitual and goal-directed behavior. Journal of Mathematical Psychology, 100, 102472. https://doi.org/10.1016/j.jmp.2020.102472

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Laís Muntini: The Environment as a Factor: Investigating the impact of the environment on cognition with VR

Abstract: Our surroundings can influence our ability to process and remember information. Factors such as ambient lighting, spatial configurations, noise, and distractions provide multimodal sensory input that affects our focus and task performance in daily life. Traditional experimental research seeks to minimize these effects by using minimalistic stimuli and laboratory environments based on two-dimensional computer screens. However, recent developments in Virtual Reality (VR) technologies have brought a paradigm shift to this research by allowing the creation and adjustment of virtual environments with control and accuracy. By mimicking natural visual perspective projection, these experiments can be tailored to investigate stimuli within context, opening new possibilities for research. We present evidence from experiments that compared the effects of the traditional mode of task presentation (2D screen) to VR (head-mounted display) on participants' performance, reaction time, and accuracy in well-established experimental psychology tasks, i.e., the Lexical Decision, Flanker, and Serial Recall tasks. All tasks had congruent results between presentation modes, pointing to the reliability of VR as a method, as findings reproduced typical patterns found in attention, memory, and word recognition research. Additionally, we varied stimuli presentation in terms of the number of dimensions (2D and 3D), level of contrast (low and high), and virtual environment conditions (rainy and sunny environment). No significant differences between 2D or 3D stimuli presentation were found. However, the effects of low contrast and rainy weather impacted performance. These results are discussed in the context of each design's limitations and chosen parameters. Altogether, this research is a step towards approximating laboratory conditions to ecological settings using VR by reproducing classical cognitive tasks in VR, integrating stimuli into a scene, and varying scene characteristics by imitating real world weather variations.

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Caroline Stankozi: Context as interactive, dynamic, and perspectival

Abstract: The role of context in cognition is more interactive, dynamic, and perspectival than commonly assumed. In my PhD project on “Cognitive Coupling”, I take a closer look at the enactive notion of ‘structural coupling’ – particularly in explanations for cognitive processes like learning.

In this short talk, I’ll give a quick overview over some central enactive ideas, like dynamic perception-action loops and sense-making (i.e., co-creating what one senses). They will enable me to emphasise why we should refrain from understanding context as something passive, static and objective. Instead, interactive processes dynamically flow throughout ever-changing agent-environment systems. Arguably, it largely depends on the agent’s situated perspective – not that of an objective observer – which context is relevant for a cognitive process.

Ekaterina Varkentin: Visual Narrative Comprehension in the Context of Aging and Stress

Abstract: Visual narrative comprehension is crucial in today's visually driven world, yet research on this ability in the context of aging remains inconclusive. Further, the interplay between aging and stress in narrative comprehension remains largely unexplored, even with stress becoming an ever-present factor in modern life. This online study examined the effects of acute stress on visual narrative comprehension in younger (N = 203, 18-57 years) and older adults (N = 212, 60-85 years). Participants were assessed under both acute stress and neutral conditions. An online stress induction tool combined mathematical and logical tasks under time pressure with elements designed to induce social stress. Participants were presented with pictorial stories comprising three panels, where the second panel was intentionally left blank. On the next page, participants judged whether the given inference for the blank panel was correct. Results showed that acute stress impaired visual comprehension and confidence in their answer in younger adults, while older adults remained unaffected. These findings suggest that with age and experience, individuals develop more differentiated event schemas, enhancing resilience to stress. The new insights provide a deeper understanding of visual narrative comprehension in the context of aging and acute stress.

Simge Hamaloğlu: Perceiving the Unseen: How Causality Impacts Accuracy Without Enhancing Metacognitive Insight

Abstract: The brain continuously constructs a coherent perception of the world by filling in missing sensory information – a process known as event completion. When an expected event is interrupted but its outcome remains causally implied, the brain bridges the gap, integrating the missing detail as if it had been perceived. This mechanism ensures fluid perception but raises the question of whether observers recognize that such details are internally generated rather than actually seen. Although research on event perception often examines accuracy and confidence, little is known about whether individuals have insight into their own judgment accuracy. Prior studies have primarily focused on ballistic motion (e.g., Papenmeier et al., 2019), such as objects moving along predictable trajectories (e.g., a thrown ball). However, everyday actions involve greater variability, making it unclear whether event completion effects generalize. To investigate this, we conducted a new experiment (N = 112) using non-ballistic stimuli to test whether these effects replicate in a different context. Using Signal Detection Theory, we calculated metacognitive efficiency (M-Ratio), a bias-free measure of the accuracy-confidence relationship. Participants watched short clips of everyday activities, followed by either a causal sequence (e.g., a person picking up an object) or a non-causal sequence (e.g., adjusting posture). They then decided whether they had seen a critical hand contact, which was removed in half of the trials, and rated their confidence. Results showed that causality reduced detection accuracy, indicating event completion. However, metacognitive efficiency remained unchanged, suggesting that while causality influences perception, it does not change awareness of these perceptual distortions, highlighting limits in evaluating one's own judgments.

Alexander Hölken: Mental Constraints - Naturalizing Mental Causation through Dynamical Systems Theory

Abstract: My doctoral thesis aims to answer the two following questions that lie at the heart of the debate about mental causation:

First, how are mental states and processes realized? My answer to this draws on the principles of Dynamical Systems Theory, which I introduce in the first section of the thesis, converging on a picture in which mental states and processes are identified with characteristic spatio-temporal patterns of brain states.

Second, how are mental states and processes, thus understood, causally efficacious? Again drawing on Dynamical Systems Theory, I argue that their causal efficacy mirrors that of higher-level constraints in other complex physical systems, such as bird flocks and ecological niches.

The dynamical theory of mental causation that I propose in my thesis conceptualizes mental states as coordinative structures within an agent-environment system that are realized by neurophysiological states and states of the agent's perceptual environment. In the final part of my thesis, I point to how this theory can be, and already is being, applied as part of empirical research paradigms within the Cognitive Sciences, with a focus on research in Sports Psychology and Social Cognition.

Joseph Petrie: Weak assertion: a decision-theoretic rational speech acts model

Abstract: TBD

Chang Liu: Investigating Speech Processing Load in a Conversational Context Task with Fish-Speech 1.5

Abstract: Spontaneous speech is rich in acoustic and contextual cues that can either aid or challenge comprehension. Yet, quantifying these variations is difficult due to the continuous nature of speech. In this thesis, I develop a novel, naturalistic experiment to investigate processing load in speech comprehension. Participants listen to recordings of real conversations and decide whether an audio segment fit the context. To model the cognitive demands of speech processing, I extracted surprisal directly from audio using a text-to-speech (TTS) model called Fish-Speech 1.5. The results show that surprisal derived from the audio itself predicts reaction times more effectively than traditional text-based surprisal. Moreover, models relying only on audio information perform as well as those incorporating both text and speech, suggesting that word-level meaning might be inferred directly from sound. These findings provide insights into human cognition and AI-driven speech modeling, hinting at the possibility of learning linguistic structure directly from speech.

Yijing (Quinn) Lin: Priming Effects on Verbal Analogy Performance Across IQ and Working Memory in Humans and AI

Abstract: Performance in verbal analogy tasks is well known to correlate with fluid intelligence, yet the underlying cognitive mechanisms remain unclear. We hypothesize that priming, prior exposure to word relationship, enhances analogy performance, and this enhancement is modulated by both IQ and age. Human participants completed a priming task followed by a set of verbal analogy problems, with intelligence independently assessed via Raven's Progressive Matrices. Our results show that priming significantly improves verbal analogy performance in individuals with lower IQ. However, among individuals with higher IQ, the priming effect is age-dependent: younger adults exhibit a reduced priming effect, whereas older adults show an enhanced effect. We propose that working memory capacity underlies these age-related differences. Specifically, with relatively higher working memory capacity, younger individuals are better able to integrate relational patterns from previous trials, thereby improving overall task performance while reducing susceptibility to priming. In contrast, older adults—with relatively lower working memory capacity—are more reliant on external cues provided by priming. We further simulated Large Language Models (LLMs) with configurations reflecting "high IQ" vs. "low IQ" and "large" vs. "small" working memory. The models exhibited similar patterns to those observed in human participants, supporting the proposed role of working memory capacity in modulating priming effects. These findings highlight the potential of LLMs as tools for investigating human cognitive performance in demanding tasks such as verbal analogy, including individual differences in fluid intelligence.

Alexandra Witt: Better together: flexibly biased learning rates in social learning

Abstract: Research on individual decision-making often finds a positivity bias, where people weight positive outcomes more strongly than negative ones during learning. This can be beneficial when rewards are rare, by amplifying relative value differences. Yet, we know very little about learning rate biases in social settings, where a key advantage is being able to vicariously learn from the negative experiences of others. This would imply a benefit for focusing on negative outcomes when learning socially, but is at odds with the seemingly inflexible positivity bias found in individual learning. Here, we examine learning rate biases across both individual and social settings, testing for adaptivity versus generally stable biases. Overall, participants appear more flexible in their learning rate biases when learning socially than what has been found in individual learning. This implies that human social learning may be more flexible and closer to normatively optimal behavior than individual learning.

Wenjia Xu: A closer look at the agent advantage effect: The Impact of Biting Orientations

Abstract: People extract and filter information from their surroundings to understand everyday events. One important component of events is event roles, which include the agent (the performer of the action) and the patient (the person or thing being acted upon) (Segalowitz, 1982). Human observers react faster toward the agent information than the patient information, which is called the agent advantage effect (Segalowitz, 1982). In our previous studies, we replicated the agent advantage effect using a search task with images of two fish, where one fish (the agent) was biting the other fish (the patient). In the present project, we investigated whether biting orientations – face-to-face (suggesting a fighting action) versus face-to-tail (suggesting a chasing action) – influence the magnitude of the agent advantage effect. We observed a significant interaction between biting orientation and search task (agent search vs patient search): the agent advantage effect emerged in the face-to-face condition but not in the face-to-tail condition. Additionally, the mean reaction time for face-to-tail conditions was significantly faster than that for face-to-face conditions. These findings suggest that the agent advantage effect depends on the context of the agent–patient relationship (fighting vs. chasing). In other words, agents are not always identified faster than patients.

Hossam Khalil: Human as an Asset - Building Human Holons for Human-Centred Production

Abstract: Although industry is increasingly considering the integration of human workers in their workflows for improved human-centred manufacturing, there is still a clear segregation of research areas. With existing research focusing either on human-centred design to facilitate assistance systems that promote frictionless workflows, mainly by addressing different aspects of the interfaces that workers use in a shop floor, or creating technologies that attempt to perceive and predict the behaviour of workers in shop floors and adapt the intended process or interaction accordingly, there is still a lack of fundamental human-centred production. One way to address this is to create human holons by modelling shop floor workers as agents, and thus being able to include them in a cyber-physical multi-agent system. This not only opens the doors for improved human-integrated workflows for different tasks such as guided assembly and human-robot handover tasks, but also allows to use human workers as the main drivers of all activities in a factory, being dependencies of different facets ranging from the creation and scheduling of production plans to the generation of robot trajectories. In this talk, an introduction to a work in progress on human holons will be presented and a framework that encapsulates these holons to address different areas of human-machine interaction (HMI) in an industrial setting and dynamically modify them to serve human comfort and satisfaction first without compromising performance will be discussed.

Maedeh Amini: Investigating Social and Emotional Cognition in Children Using Face Processing in EEG Studies: A Systematic Review

Abstract: TBD

Hassane Kissane: Neural and computational Modeling of Verb-Particle combinations

Abstract: This study investigates the internal representations of verb-particle combinations, called multi-word verbs, within transformer-based large language models (LLMs), specifically examining how these models capture lexical and syntactic properties at different neural network layers. Using the BERT architecture, we analyze the representations of its layers for two different verb-particle constructions: phrasal verbs like give up and prepositional verbs like look at. Our methodology includes training probing classifiers on the model output to classify these categories at both word and sentence levels. The results indicate that the model's middle layers achieve the highest classification accuracies. To further analyze the nature of these distinctions, we conduct a data separability test using the Generalized Discrimination Value (GDV). While GDV results show weak linear separability between the two verb types, probing classifiers still achieve high accuracy, suggesting that representations of these linguistic categories may be non-linearly separable. This aligns with previous research indicating that linguistic distinctions in neural networks are not always encoded in a linearly separable manner. These findings computationally support usage-based claims on the representation of verb-particle constructions and highlight the complex interaction between neural network architectures and linguistic structures.

Max Mittenbühler: A Rational Trade-Off Between the Costs and Benefits of Automatic and Controlled Processing

Abstract: Humans seem to arbitrate between automatic and controlled processing by optimizing a trade-off between cognitive effort and performance. Previous research has described ways of how these costs and benefits can be quantified and how the trade-off between them can be performed. However, it remains unclear how the costs should be weighed relative to the benefits and how the cost of the arbitration mechanism itself factors in. In this talk, I will show how abstract context representations can provide an efficient and near-optimal solution to this meta-control problem. As a proof of concept, I will demonstrate that the congruency and proportion congruency effects in the Stroop task directly result from this, given an environment with specific statistical regularities.